

# Microbes in Human Welfare

## 1. During conversion of milk into curd

- (1) Lactic acid bacteria completely digest milk proteins
- (2) LAB produce acids that coagulate milk
- (3) Vitamin B12 content decreases
- (4) Some unicellular eukaryotes are involved

## 2. Statins are

- a) Commercial blood-cholesterol lowering agents.
- b) Produced by yeast.
- c) Competitive inhibitors of enzyme responsible for cholesterol synthesis.

Choose the correct one(s).

- (1) a & b only
- (2) b & c only
- (3) a & c only
- (4) All a, b & c

## 3. How many of the following alcoholic beverages are produced with the help of distillation?

Wine, Brandy, Beer, Rum, Whisky, Vodka

- (1) 2
- (2) 3
- (3) 4
- (4) 5

## 4. Lipases are commercially obtained from

- (1) *Aspergillus niger*
- (2) *Penicillium notatum*
- (3) *Candida lipolytica*
- (4) *Streptococcus pneumoniae*

## 5. The main gas produced during beverage production, cheese making and dough fermentation is

- (1)  $O_2$
- (2)  $H_2S$
- (3)  $CO_2$
- (4)  $CH_4$

## 6. Read the given statements

**Statement A: Ministry of Environment and Forest has initiated Ganga Action Plan.**

**Statement B: Effluent from secondary treatment cannot be released into water bodies.**

Choose the most appropriate answer from the options given below:

- (1) Only A is correct
- (2) Only B is correct
- (3) Both A & B are correct
- (4) Both A & B are incorrect

## 7. Match List-I with List-II.

- | List-I           | List-II                                 |
|------------------|-----------------------------------------|
| (a) Swiss cheese | (i) <i>Caryota urens</i>                |
| (b) Toddy        | (ii) <i>Propionibacterium sharmanii</i> |
| (c) Wine         | (iii) <i>Aspergillus</i>                |
| (d) Pectinases   | (iv) <i>Saccharomyces</i>               |

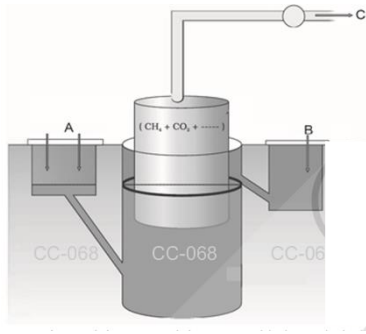
Select the correct answer from the following options.

- (1) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- (2) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
- (3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (4) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

## 8. Antibiotics, which are 'pro-life' with reference to human beings

- (1) Were first discovered while working on *Streptococcus*.
- (2) Are known to retard the growth of disease-causing microbes.
- (3) Are regarded as the most significant discoveries of twenty first century
- (4) Were used to treat soldiers and they were first discovered by Ernest Chain and Howard Florey.

9. The given diagram is of biogas plant with label A, B and C.



A → Slurry of dung containing anaerobic bacteria is fed here

B → Cattle dung for anaerobic fermentation is fed here.

C → Biogas obtained from anaerobic digestion is collected from here.

Which of the above labels is/are correctly described?

- (1) A and B only
- (2) C only
- (3) A and C only
- (4) B and C only

10. Which of the following statement is incorrect w.r.t. *Trichoderma*?

- (1) They are used to produce immunosuppressive agents for organ transplant patients.
- (2) They are biocontrol agents, being developed for use in the biological treatment of plant disease.
- (3) They are free living fungi that are common in root ecosystem of the plants.
- (4) They are effective in the control of butterfly caterpillars.

11. Which is wrong statement for methanogen bacteria:

- (1) They produce  $\text{CH}_4$ ,  $\text{CO}_2$  and  $\text{H}_2$
- (2) They are found in rumen of cattles
- (3) They breakdown the cellulose
- (4) They digest cellulose in human body

12. Read the following statements :

- (A) Cyclosporin A is used as a clot buster
- (B) Whisky, brandy and rum are produced by distillation of fermented broth
- (C) NPV are the major Baculoviruses used as biological control agents
- (D) The biogas production technology in India is developed by ICAR

Which of the above statements are not incorrect?

- (1) A & B
- (2) B & C
- (3) C & D
- (4) A & D

13. Mycorrhizal association does not

- (1) Help to absorb phosphorus from soil
- (2) Help plants to show resistance to root-borne pathogens
- (3) Increases tolerance of plants to salinity
- (4) Retard plant growth and development

14. *Trichoderma* species are beneficial to human beings in various ways. Which among the given is not a benefit provided by *Trichoderma*?

- (1) Effective biocontrol agents of several plant pathogens
- (2) Produce immunosuppressive agent
- (3) Remove clots from blood vessels of patients that have undergone myocardial infarction
- (4) Used to obtain cyclosporin A

15. Ladybird beetles are used in the biocontrol of

- (1) Mosquitoes
- (2) Aphids
- (3) Plant pathogens
- (4) Caterpillars

16. Select the incorrectly matched pair

- (1) Curd – *Lactobacillus*
- (2) Bread – *Saccharomyces*
- (3) Dosa – *Leuconostoc*
- (4) Citric acid – *Acetobacter*

**17. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R) .**

**Assertion (A) : Jatropha is a plant used as a source of hydrocarbon to produce petroleum.**

**Reason (R) : It is a plant, produces milk like substance latex in which hydrocarbons are present**

**In the light of the above statements, choose the most appropriate answer from the options given below :**

- (1) If (A) & (R) both are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is incorrect.
- (3) (A) is not correct but (R) is correct.
- (4) If (A) & (R) both are correct and (R) is the correct explanation of (A).

**18. Which one of the following is an example of carrying out biological control of pests / diseases using microbes :**

- (1) Bt-cotton to increase cotton yield
- (2) Lady bird beetle against aphids in mustard
- (3) Trichoderma sp. against certain plant pathogens
- (4) Nucleopolyhedrovirus against white rust in Brassica

**19. In the biological treatment of sewage, flocs obtained are masses of**

- (1) Anaerobic bacteria
- (2) Fungal filaments only
- (3) Aerobic bacteria only
- (4) Bacteria associated with fungal filaments

**20. During treatment of waste water, sequential filtration of floating debris and sedimentation steps are included under**

- (1) Primary treatment
- (2) Physico chemical treatment
- (3) Secondary treatment
- (4) Tertiary treatment

**21. Which of the following bacteria is associated with biofertilizer :**

- (1) Azotobacter
- (2) Methanobacterium
- (3) Agrobacterium
- (4) Trichoderma

**22. Given below are two statements:**

**Statement-I : A bacterium Propionibacterium sharmanii is used to produce more CO<sub>2</sub> in Swiss cheese.**

**Statement-II : Saccharomyces cerevisiae is used to make breads spongy due to much production of CO<sub>2</sub> .**

**In the light of the above statements, choose the most appropriate answer from the options given below:**

- (1) Both Statement-I and II are correct.
- (2) Statement-I is correct but statement-II is incorrect.
- (3) Statement-I is incorrect and Statement-II is correct.
- (4) Both Statement-I and Statement-II are incorrect.

**23. Bioactive molecule.....is used as immunosuppressive agent in organ transplant patients. It is produced by the fungus.....**

- (1) Antibiotic, Trichoderma polysporum
- (2) Cyclosporin A, Trichoderma polysporum
- (3) Trichoderma polysporum, Cyclosporin A
- (4) Trichoderma polysporum, Antibiotic

**24. Mark the wrong statement.**

- (1) Puffed up appearance of dough is due to CO<sub>2</sub> released by alcoholic fermentation of food
- (2) Toddy is prepared by fermenting sap of palm by microbes
- (3) Microbes are also used to ferment fish and bamboo shoots to make foods
- (4) Cheese are ripened by some specific multicellular microbes only

**25. All solids that settle during primary treatment of waste water is called**

- (1) Primary sludge
- (2) Primary effluent
- (3) Flocs
- (4) Activated sludge

**26. All of the following are related to secondary treatment of sewage, except**

- (1) Production of methane
- (2) Digestion of sludge
- (3) Killing of pathogens using UV rays
- (4) Reduction in BOD of water

**27. Roquefort cheese are ripened by following specific organisms which gives a particular flavour**

- (1) Bacteria
- (2) Actinomycetes
- (3) Cyanobacteria
- (4) Fungi

**28. Identify the household product which is prepared by fermentation by bacteria?**

- (1) Dosa
- (2) Bread
- (3) Camembert cheese
- (4) Beer

**29. Biocontrol agents are preferred over chemical methods of pests and disease control because**

- a. They are non toxic for non targeted organisms**
- b. They do not kill useful organisms**
- c. They keep pests at manageable levels**

- (1) Only a
- (2) Only b
- (3) Only b and c
- (4) All a, b and c

**30. Select the odd one w.r.t. biofertilisers.**

- (1) Anabaena
- (2) Bacillus thuringiensis
- (3) Azotobacter
- (4) Azospirillum

**31. Choose the odd one w.r.t. beverages produced by distillation of fermented broth.**

- (1) Whisky
- (2) Brandy
- (3) Rum
- (4) Beer

**32. Organic acids have been proved to be useful for human beings. Examples of some acid producers are given below.**

**Select odd one w.r.t. prokaryotic acid producer**

- (1) Acetobacter aceti
- (2) Saccharomyces cerevisiae
- (3) Lactobacillus
- (4) Clostridium butylicum

**33. Penicillin was discovered by Alexander Fleming, while working on**

- (1) Rhizopus stolonifer
- (2) Streptococci
- (3) Clostridium butylicum
- (4) Staphylococci

**34. Read the given statements of Assertion and Reason.**

**Assertion: Microbes play a major role in treating millions of gallons of waste water everyday across the globe.**

**Reason: No man-made technology has been able to rival the microbial treatment of sewage.**

**In the light of the above statements, choose the most appropriate answer from the options given below:**

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false statement
- (4) Both Assertion & Reason are false statements



**35. Which of the following statements is incorrect?**

- (1) Biogas can be used as source of energy as it is inflammable
- (2) Bacteria which grow anaerobically on cellulosic material produce methane, CO<sub>2</sub> and H<sub>2</sub>
- (3) Methanogens are found in aerobic sludge during sewage treatment
- (4) Dung of cattle is used for generating gobar gas

**36. Read the following statements and state true (T) or false (F).**

**A. Dragonflies are useful to get rid of mosquitoes.**

**B. The ministry of environment and forests has developed the technology of biogas production in India.**

- (1) A-(F), B-(T)
- (2) A-(T), B-(F)
- (3) A-(F), B-(F)
- (4) A-(T), B-(T)

**37. We can use antibiotics for cure of all of the following diseases, except**

- (1) Diphtheria
- (2) Plague
- (3) Leprosy
- (4) Influenza

**38. The enzyme used in detergent formulation to remove oily stains is**

- (1) Lipase
- (2) Pectinase
- (3) Streptokinase
- (4) Cellulase

**39. Select the incorrect one for the main aims of organic farming.**

- (1) Usage of biofertilisers
- (2) Significantly reduces soil pollution
- (3) Discourage usage of chemical fertilisers
- (4) Promotes conservation of beneficial insects of an area and discourage use of insecticides

**40. Read the following Assertion (A) and Reason (R) statements and choose the correct option.**

**Assertion (A) : Majority of baculoviruses in genus Nucleopolyhedrovirus are excellent candidates for IPM (Integrated Pest Management).**

**Reason (R) : Baculoviruses are of broad spectrum insecticidal application.**

- (1) Both Assertion & Reason are true and reason is the correct explanation of assertion
- (2) Both Assertion & Reason are true but reason is not the correct explanation of assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

**41. What role is played by lactic acid bacteria (LAB) in our stomach ?**

- (1) Beneficial
- (2) Harmful
- (3) Neutral
- (4) Rarely beneficial

**42. Statins commercially used as blood cholesterol lowering agents are produced by :**

- (1) Yeast
- (2) Algae
- (3) Bacteria
- (4) Actinomycetes

**43. Identify the correct set of statements out of followings :**

- (a) Commercial production of citric acid is done in the presence of *Aspergillus niger* fungi
- (b) First antibiotic was penicillin produced from a fungi
- (c) *Trichoderma* is a free living fungi which is common in the root ecosystem
- (d) Streptokinase is obtained from a bacterium called *Streptobacillus*

**Choose correct answer**

- (1) a, b, c and d
- (2) only a and b
- (3) a, b and c
- (4) b, c and d

**44. Identify the blank spaces A, B, C and D in the following given table and select the correct options :**

| Type of microbe | Scientific name        | Commercial product |
|-----------------|------------------------|--------------------|
| Bacterium       | A                      | Streptokinase      |
| B               | Aspergillus niger      | Citric acid        |
| Fungus          | Trichoderma polysporum | C                  |
| Bacterium       | D                      | Butyric acid       |

- (1) A-Streptococcus. B-Fungus, C-Cyclosporin-A, D-Clostridium butylicum  
 (2) A-Clostridium butylicum, B-Streptococcus. C-Fungus. D-Cyclosporin-A  
 (3) A-Sfrepococcus, B-Yeast, C-Cyclosporin-A, D-Lactobacillus  
 (4) A-Streptococcus, B -Cyclosporin-A, C-Sratins, D-Clostridium butylicum

**45. In sewage treatment activated sludge**

- (1)Serves as inoculum in aeration tank  
 (2)Is produced during primary treatment  
 (3)Contains soil & small pebbles  
 (4)Is digested during tertiary treatment

**46. Secondary treatment of sewage does not include**

- (1)Activity of heterotrophic bacteria  
 (2)Activity of anaerobic bacteria  
 (3)Biogas production  
 (4)Chemical treatment of sewage to reduce its BOD

**47. Given below are two statements:**

**Statement-I : Cyclosporine-A is a bioactive molecule used as an immune suppressive drug during organ transplantation.**

**Statement-II : Statins produced from yeast, is used as blood cholesterol lowering agents.**

**In the light of the above statements, choose the most appropriate answer from the options given below:**

- (1) Both Statement-I and II are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both Statement-I and Statement-II are incorrect.

**48. Which of the following is correct ?**

- (1) VAM is not a biofertilizer  
 (2) Azolla and BGA are the not biofertilizers  
 (3) Synthetic fertilizers cause pollution  
 (4) Probiotics are live pathogenic food supplement

**49. Propionibacterium sharmanii is a prokaryotes and it is used for the production of**

- (1) Toddy  
 (2) Swiss cheese  
 (3) Clot buster  
 (4) Biogas

**50. Match the following list of microbes and their importance**

|     |                                |       |                                                 |
|-----|--------------------------------|-------|-------------------------------------------------|
| (a) | Saccharomyces cerevisiae       | (i)   | Production of immuno suppressive agents         |
| (b) | Monascus purpureus             | (ii)  | Ripening of swiss cheese                        |
| (c) | Trichoderma Polysporum         | (iii) | Commercial production of ethanol                |
| (d) | Propioni - bacterium sharmanii | (iv)  | Production of blood cholesterol lowering agents |

- (1) a-(iii), b-(i), c-(iv), d-(ii)  
 (2) a-(iii), b-(iv), c-(i), d-(ii)  
 (3) a-(iv), b-(iii), c-(ii), d-(i)  
 (4) a-(iv), b-(ii), c-(i), d-(iii)

# Answer Key

1. (2)
2. (4)
3. (3)
4. (3)
5. (3)
6. (1)
7. (1)
8. (2)
9. (3)
10. (4)
11. (4)
12. (2)
13. (4)
14. (3)
15. (2)
16. (4)
17. (4)
18. (3)
19. (4)
20. (1)
21. (1)
22. (1)
23. (2)
24. (4)
25. (1)
26. (3)
27. (4)
28. (1)
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30. (2)
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34. (2)
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36. (2)
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38. (1)
39. (4)
40. (3)
41. (1)
42. (1)
43. (3)
44. (1)
45. (1)
46. (4)



- 47. (1)
- 48. (3)
- 49. (2)
- 50. (2)

